Lecture 3 – AERM Macro

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Income Processes in Model and Data

- Dynamic model should have a realistic income process
- Huge literature
  - Persistent-transitory model with random walk and stabilizing force (Gottschalk-Moffitt)
  - Growth rate heterogeneity (see discussion in Guvenen (2007))
  - Rich age profiles with endg. human capital accumulation (Hugget et al (2011))
  - Non-normal innovations (Guvenen et al)
  - Innovations that change over the business cycle (Guvenen et al)
- We focus on simple AR(1) process

Income Processes in Model and Data

- Log income depends on past income and innovation u
- Say $u \sim N(0, \sigma_u)$

$$y_{i,t+1} = \rho y_{i,t} + u_{i,t+1} \tag{1}$$

- Can you show that the stationary distribution is mean-zero with variance $\frac{\sigma_u^2}{1-\rho^2}$?
- Problem: how do we map continous income process into discrete global maximization routine?

Rouwenhorst Method

- Rouwenhorts (1995) approximates stochastic process using discrete states
- See Kopecky/Suen (2009)
- N-state Markov chain with
  - Symmetric and evenly spaced state space centered around zero
  - Largest log income ψ
  - Transition matrix Θ
  - Θ only depends on two parameters, p and q

Rouwenhorst Method: Recursive approach

- Construct a "big" transition matrix recursively

1. For $N = 2$ we start with $\Theta_2 = \begin{bmatrix} p & 1-p \\ 1-q & q \end{bmatrix}$
2. For $N \geq 3$ use recursion

$$\Theta_N = p \begin{bmatrix} \Theta_{N-1} & \mathbf{0} \\ \mathbf{0}' & 0 \end{bmatrix} + (1-p) \begin{bmatrix} \mathbf{0} & \Theta_{N-1} \\ 0 & \mathbf{0}' \end{bmatrix} + (1-q) \begin{bmatrix} \mathbf{0}' & 0 \\ \Theta_{N-1} & \mathbf{0} \end{bmatrix} + q \begin{bmatrix} 0 & \mathbf{0}' \\ \mathbf{0} & \Theta_{N-1} \end{bmatrix}$$

where $\mathbf{0}$ is a $N-1$ column vector of zeros

- Divide every row excet top and bottom in Θ_n by 2 after each iteration

Rouwenhorst Method excellent approximation to AR(1)

- Let $p = q = (\rho + 1)/2$
- Let $\psi = \sqrt{\frac{\sigma_u^2}{1-\rho^2} \cdot (N-1)}$
- Could derive properties of stochastic process theoretically (Kopecky/Suen, 2009)
  - Key takeway: Tauchen-method bad for $\rho \approx 1$, Rouwenhorst still works!
- Use simulations to show

$$\mathbb{E}\left[y_{t+1} \middle| y_t\right] = \rho y_t$$

$$\mathbb{E}\left[y\right] = 0$$

$$Var\left[y\right] = \frac{\sigma_u^2}{1-\rho^2}$$

Simulating an income process

- Suppose we have settled for some transition matrix $\Pi\left(\rho, \sigma_u^2, N\right)$
  - We focus on simple case in this class where Π is exogenous
  - Exciting topics where this is not the case
    - Labor markert frictions, unemployment, on the job search, ...
    - Endogenous human capital accumulation
- How do you actually simulate the model?

Simulating an income process: simple example

- To build intuition, suppose you want to simulate coin flips where the probabillity of tails is 60%
- You want to simulate one hundered coin flips
- You lost the (biased) coin, but you do know: $\mathbb{P}(tails) = .6$
- You are in luck: I give you a vector of random numbers $\mathbf{x} = [x_1, x_2, ..., x_{100}]'$ that are iid $\sim U[0, 1]$
- Rule: if $x_n < .6 \rightarrow n^{th}$ toss is tails, otherwise heads
- Claim: this vector allows you to simulate the coin flip, even though you lost the coin
- What if the coin was fair, which threshold would you have picked?

Simulating an income process using CDF and uniform draws

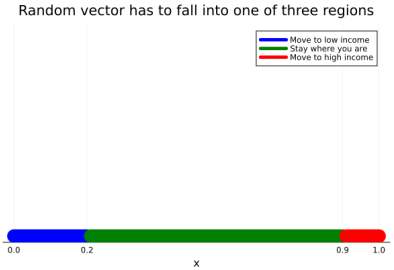
- Suppose we start out with some 1000 households (unit i)
- Suppose we want to simulate them over a ten year period (unit T)
  - $N_sim = 1000$, $T_sim = 10$
- Suppose we want to draw initial income at time 1 using the stationary distribution

Simulating an income process

- Step 0: Generate a random matrix of independent draws from $U[0, 1]$ of size $N_sim \times T_sim$
  - Disclaimer: can there be a truly random number sequence generated by a computer?
  - Quasi-random good enough for all statistical purposes
- Step 1: start out with an initial draw at time zero
  - Let's say $y_{i,0} = 0 \quad \forall i$

Simulating an income process: richer income states

- Suppose income can take three values $y \in \{y_{\mathrm{L}}, y_{\mathrm{M}}, y_{\mathrm{H}}\}$
  - Focus on conditional distribution of household in the middle: $y_t = y_{\mathrm{M}}$
  - $p(y_{\mathrm{L}}|y_{\mathrm{M}}) = .2, p(y_{\mathrm{M}}|y_{\mathrm{M}}) = .7, p(y_{\mathrm{H}}|y_{\mathrm{M}}) = .1$
- I again give you a random draw x from a uniform distribution



- **All we need is the cdf** to use uniform draws to simulate anything!

Simulating an income process: back to step 2

- Step 2.1: Do we have the cdf? Can be easily computed from Π!
- Step 2.2: Run forward in time for each individual – recursive

```
# and now you can simulate the rest of the income path for each individual
inc_mat = zeros(N_sim,T_sim)
pos_mat = zeros(N_sim,T_sim)
for i = 1:N_sim
    y_vec = sol_rouwen.y_vec[Int(pos_N_initial[i])]*ones(T_sim)
    y_pos = Int.(pos_N_initial[i]*ones(T_sim))
    for t=2:T_sim
        y_pos[t] = findfirst(x -> x >= rand_mat[i,t], Π_cdf[:,Int(y_pos[t-1])])
        y_vec[t] = sol_rouwen.y_vec[Int(y_pos[t])]
    end
    inc_mat[i,:] = y_vec
    pos_mat[i,:] = y_pos
end
```

Simulating an income process: we did it!

- We now have a big data matrix with N rows and T columns
- This is simulated data, but we treat it like the actual data (this is the whole point)
- You will code this up and then you get to run a regression to tell me
  - What the autoregressive coefficient $\hat{\rho}$ is
  - Whether the 95% confidence interval covers the true ρ

Big Deal because now you can simulate your model!

- You already know how to solve for the policy function $c(B, y)$
- Given an initial distribution, you can
  - Simulate a households' joint evolution of income and consumption $\{c_{i,t}, y_{i,t}\}$
  - Ask questions like: what is the passthrough from inocme shocks to consumption on average in model
  - How does this compare to the data – run the same regression, very neat!

Income and Consumption Dynamics

Link between Income and Consumption

- Classic question: how do income shocks map into consumption adjustment?
  - Macro-focus: marginal propensity to consume (MPC) matters for fiscal and monetary policy
  - Micro-focus: how much insurance is there against income shocks
    - Perfect insurance – no need to provide government relief for distressed households
    - No insurance – different debate

How much insurance is there for income shocks

- Follow Blundell/Pistaferri/Preston (AER, 2008)
- Canonical model of income and consumption
- Things we ignore for this:
  - Richer income processes – exciting research topic
  - Endogenous income processes – exciting research topic
  - GE – matters for everything so non-trivial simplification (people may disagree)
  - Richer models of consumption (non-homotheticities) – exciting research topic

Income processes in BPP

- Simplest version is persistent-transitory workhorse model
- Annual frequency – log income y

$$y_t = p_t + \nu_t \tag{2}$$

$$p_t = p_{t-1} + \zeta_t, \quad \zeta_t \sim_{iid} N\left(0, \sigma_p^2\right) \tag{3}$$

$$\nu_t = \epsilon_t, \quad \epsilon_t \sim_{iid} N\left(0, \sigma_\nu^2\right) \tag{4}$$

  - Note this income process is inconsistent with inf. horizon + stationary dist'n
    - Shocks are too persistent so distribution would be fanning out forever
    - But works well for life cycle problems, which is the focus here
    - Works well on finite grid, which is what we will use

Excursion: income inequality across cohorts in the USA

- Cohort = Time - Age, then run: $stat_{age,t} = \alpha + \beta_c + \beta_t + \delta_{age} + u_{age,t}$
- Fanning out suggests persistent shocks

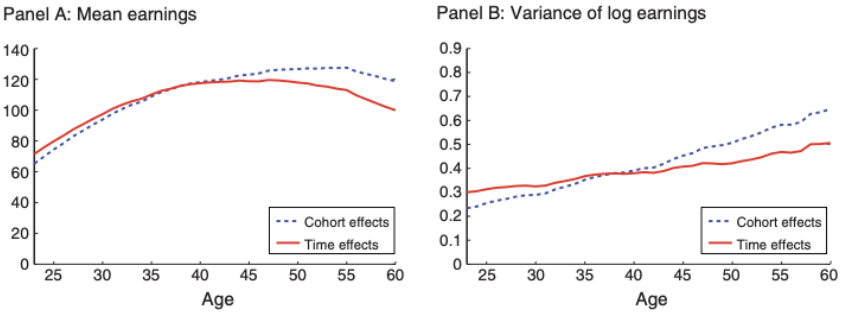


Figure 1: Hugget et al (2011)

Passthrough from income to consumption

- In principal, can run regression of the following sort (Townsend, 1994)

$$c_{i,t} = \beta y_{i,t} + \gamma' \mathbf{x}_{i,t} + u_{i,t}$$

- Obs: we would think that response quite different for shocks to permanent (ζ) and transitory (ϵ) component of income
- Problem: we only see y but we never see individual components of p and ν
- Solution: impose some structure

  (but not a lot – this is why people like the BPP paper so much)

BPP uses income growth rates to estimate income process (side show)

- First stage regression to take out observables and age
- Look at first difference of log $\implies$ growth rate

$$\Delta y_{t+1} = y_{t+1} - y_t \tag{5}$$

$$= \zeta_{t+1} + \Delta\epsilon_{t+1} \tag{6}$$

- Can you show the following:

$$\mathbb{E}_i\left[\Delta y_t \Delta y_t\right] = \sigma_p^2 + 2\sigma_\epsilon^2$$

$$\mathbb{E}_i\left[\Delta y_{t+1} \Delta y_t\right] = -\sigma_\epsilon^2$$

$$\mathbb{E}_i\left[\Delta y_{t+2} \Delta y_t\right] = 0$$

BPP models consumption response (main deal)

- Approximate consumption response based on permanent income hypothesis
- Forward-looking households consume a fraction of their financial and human wealth
  - Human wealth is the present discounted value of expected labor income
- Add. assumption for approximation to be exactly right
  - Quadratic utility (shut down precautionary savings)
  - No borrowing constraint (ensure unconstrained household)
  - Very low interest rate (ensure main source of income is human wealth)
  - Turns out to be a good approixmation
  - I recommend Kaplan/Violante (AEJ: Macro, 2010) – extremely insightful

BPP derive consumption response in growth rates

- Tack on error u
- Derive equation of consumption growth as a function of shocks

$$\Delta c_{i,t} = \phi \zeta_{i,t} + \psi \epsilon_{i,t} + u_{i,t} \tag{7}$$

- Which "passthrough" coefficient (ϕ, ψ) you think will be larger in theory?
  - Recall ζ pertains to persistent shocks that permanently shift your labor income
  - In contrast, ϵ is a mean reverting transitory shock

Identification of passthrough

- BPP show identification using "instruments" for shocks $\Delta\zeta$ and $\Delta\epsilon$

$$\phi = \frac{\mathbb{E}\left[\Delta c_{i,t}(\Delta y_{i,t-1} + \Delta y_{i,t} + \Delta y_{i,t+1})\right]}{\mathbb{E}\left[\Delta y_{i,t}(\Delta y_{i,t-1} + \Delta y_{i,t} + \Delta y_{i,t+1})\right]} \tag{8}$$

$$\psi = \frac{\mathbb{E}\left[\Delta c_{i,t}\Delta y_{i,t+1}\right]}{\mathbb{E}\left[\Delta y_{i,t}\Delta y_{i,t+1}\right]} \tag{9}$$

- Amazing: given inc. & cons. data, you can estimate "partial insurance" coefficients $1 - \phi$

Dependence on model

- Borrowing constraint – you will explore this issue in the problem set
- Income process – you will explore this issue in the problem set
- Other issues:
  - Advance information
- Also, not necessarily structural object
  - Baked in endg. labor supply response (observed)
  - Baked in gov. transfer (observed)
  - Baked in informal networks (unobserved)
  - Measurement error problem for trans. shock

Results BPP

TABLE 7—MINIMUM-DISTANCE PARTIAL INSURANCE AND VARIANCE ESTIMATES

Consumption: Income: Sample:	Nondurable net income baseline	Nondurable earnings only baseline	Nondurable male earnings baseline
ϕ	0.6423	0.3100	0.2245
(Partial insurance perm. shock)	(0.0945)	(0.0574)	(0.0493)
ψ	0.0533	0.0633	0.0502
(Partial insurance trans. shock)	(0.0435)	(0.0309)	(0.0294)

Notes: This table reports DWMD results of the parameters of interest. We also estimate time-varying variances of measurement error in consumption (results not reported for brevity). See the main text for details. Standard errors in parentheses.

Figure 2: Blundell/Pistaferri/Preston (2008)

Own Research & Repetition

- Repetition
- Income Risk and Consumption Volatility in Rural-Urban Environments & Agg. Development

References

Guvenen, F. (2007). Learning your earning: Are labor income shocks really very persistent? *American Economic Review 97*(3), 687–712.